Predicting ICU Mortality Risk Using Machine Learning Models

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Github: https://github.com/1171958037-creator/ICU-mortality-prediction

1. Introduction

Early identification of critically ill patients with a high risk of mortality is an important challenge in intensive care medicine. Predictive models based on electronic health records can support clinical decision-making by identifying patients who may require closer monitoring or early intervention.

In this study, clinical variables from the MIMIC dataset were used to develop models for predicting mortality risk among ICU patients. The dataset contains demographic information, laboratory test results, and physiological indicators recorded during ICU stays, making it suitable for machine learning analysis.

To identify the most informative predictors, L1-regularized logistic regression was used as an embedded feature selection method. Several machine learning models were then implemented and compared, including Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). Model performance was primarily evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC), which is commonly used when working with imbalanced clinical datasets.

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2. Data Preprocessing

Before model training, several preprocessing steps were performed to improve data quality.

The original dataset contained 65,366 observations and 140 variables. Examination of missing values showed that more than 80 variables had missing rates above 50%. These variables were removed because such high levels of missingness make them difficult to use reliably in predictive modeling.

The outcome variable in this study was hospital_expire_flag, indicating whether the patient died during hospitalization. To avoid potential data leakage, the variable icu_death_flag, which is closely related to the outcome, was removed.

Additional variables were also excluded: identifier variables (subject_id, hadm_id, stay_id), ICU unit information, time-related variables such as admission and discharge times. These variables do not provide meaningful information for predicting mortality risk.

For the remaining features, missing values were handled using simple imputation. Numerical variables were filled using the median, while categorical variables were filled using the mode. Categorical variables such as race and gender were converted into numerical form using one-hot encoding.

After preprocessing, the final dataset contained 65,366 samples and 101 features.

The dataset was then split into training and testing sets using an 80/20 split, with stratified sampling to maintain the class distribution. Feature scaling was performed using StandardScaler to ensure comparable variable scales.

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3. Feature Selection

Feature selection was performed using L1-regularized logistic regression with cross-validation (LogisticRegressionCV). L1 regularization encourages sparsity by shrinking the coefficients of less informative variables toward zero. As a result, only the most relevant predictors are retained.

Cross-validation was used to automatically determine the optimal regularization strength. To improve computational efficiency, the parameter n_jobs = −1 was used to enable parallel processing.

After training the model, the optimal regularization parameter selected by cross-validation was C = 0.0001, indicating relatively strong regularization. Under this setting, only three variables remained with non-zero coefficients:

Feature	Coefficient
SpO_2 min (minimum oxygen saturation)	−0.0027
GCS motor (Glasgow Coma Scale motor score)	−0.0778
GCS eyes (Glasgow Coma Scale eye response)	−0.0114

Table 1: Selected features and coefficient values

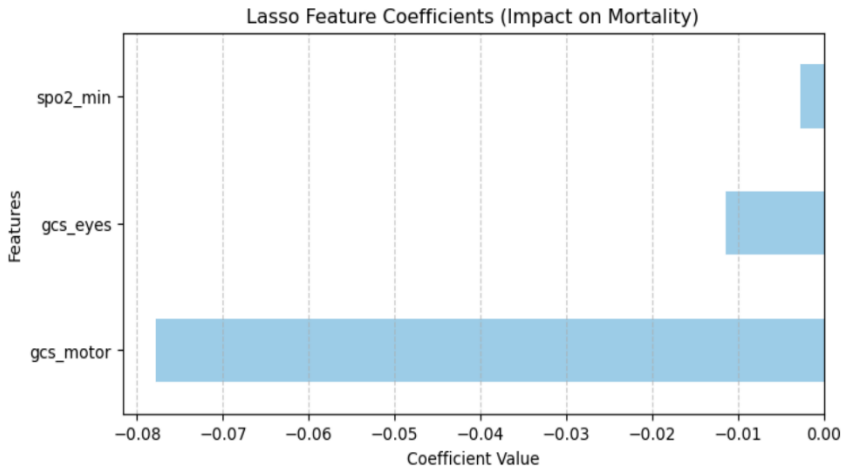

Figure 1: Lasso Feature Coefficient (Impact on Mortality)

These variables represent important indicators of respiratory and neurological status. Other clinical measurements such as blood pressure or heart rate may provide similar information but could be strongly correlated with these variables, leading the L1 regularization process to retain only the most representative features.

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4. Predictive Models

Four machine learning models were implemented and compared: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM)

Logistic Regression served as a baseline linear model. Decision Trees can capture nonlinear decision rules, while Random Forest improves stability by combining multiple decision trees. SVM attempts to find an optimal separating boundary between classes.

To maintain clinical interpretability and prevent overfitting on the noisy MIMIC-IV dataset, specific hyperparameters were strategically selected for each model. The Decision Tree was restricted to a maximum depth of 5 to ensure simple, clinically relevant rules, while the Random Forest utilized an ensemble of 100 trees to reduce prediction variance and enhance stability across heterogeneous patient profiles. For the SVM, a linear kernel was chosen to provide a clear decision boundary and direct coefficient insights suitable for a low-dimensional 3-feature space, avoiding the "black-box" complexity of non-linear kernels. Crucially, the class_weight='balanced' parameter was applied to all models to automatically adjust for the severe class imbalance between survivors and mortality cases, thereby ensuring that the algorithms remain sensitive to high-risk patients rather than biasing toward the majority class.

All models were trained using the three selected features obtained from the feature selection stage.

5. Results

5.1 Model Performance (AUC)

Model performance was evaluated using AUC.

Model	AUC
Logistic Regression	0.770
Decision Tree	0.780
Random Forest	0.761
SVM	0.769
Logistic Regression (all features)	0.868

Table 2: AUC scores of different models

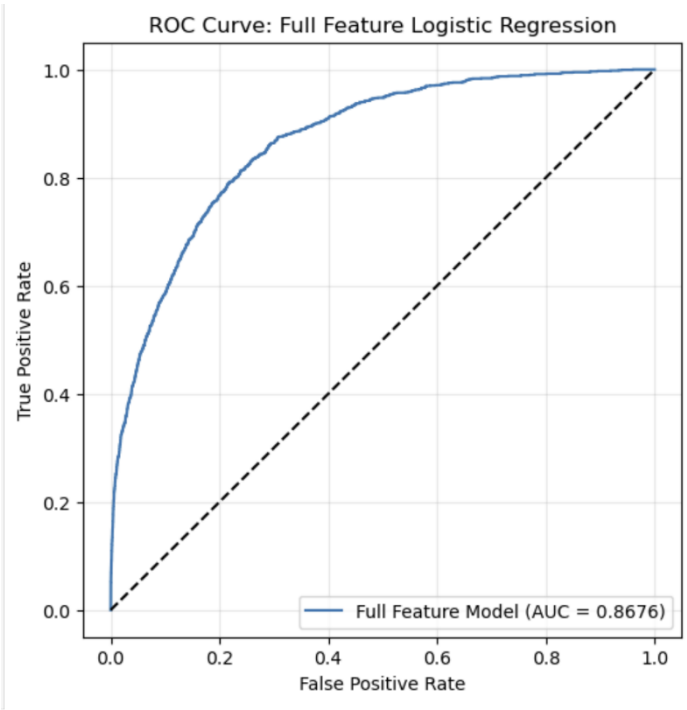


Figure 2: ROC Curve: Full Feature Logistic Regression

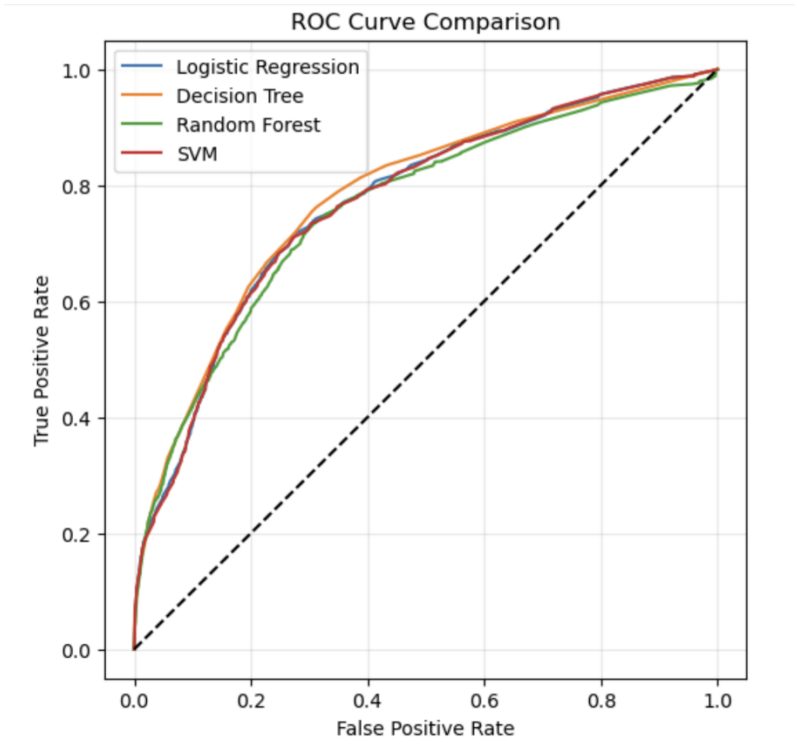


Figure 3: ROC Curve Comparison (Logistic Regression, Decision Tree, Random Forest, SVM)

Among the models using the selected features, the Decision Tree achieved the highest AUC (0.780). However, the performance differences between the models were relatively small.

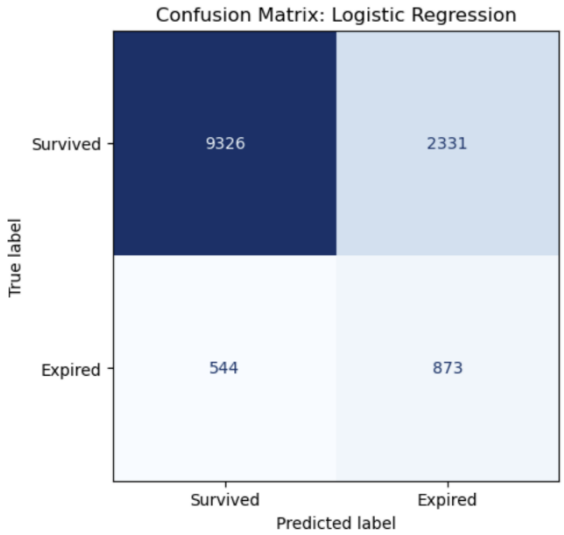
For comparison, Logistic Regression trained on the full feature set achieved a higher AUC of 0.868, suggesting that additional variables still contain useful predictive information.

5.2 Classification Performance

Classification reports showed differences in how well each model identified mortality cases.

For example, the Decision Tree model achieved the highest recall for expired patients (0.72), indicating better sensitivity in detecting high-risk patients. In contrast, Logistic Regression and SVM showed slightly lower recall but higher precision.

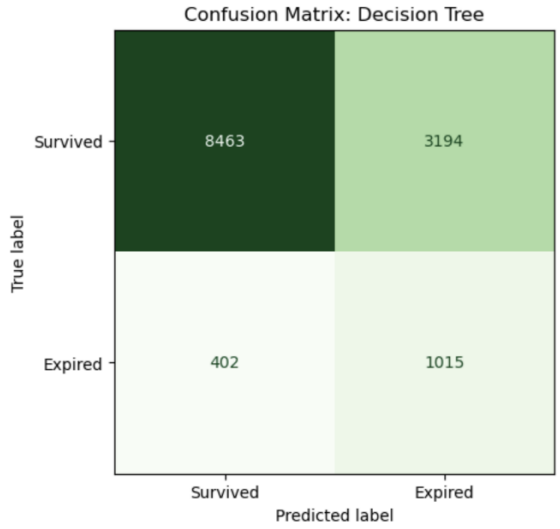
This result highlights an important trade-off. While some models achieve higher overall accuracy, models with higher recall for mortality cases may be more valuable in clinical settings, where missing high-risk patients can have serious consequences.




```
--- Logistic Regression Classification Report ---
              precision    recall  f1-score   support

    Survived       0.94      0.80      0.87     11657
     Expired       0.27      0.62      0.38      1417

    accuracy                           0.78     13074
   macro avg       0.61      0.71      0.62     13074
weighted avg       0.87      0.78      0.81     13074
```

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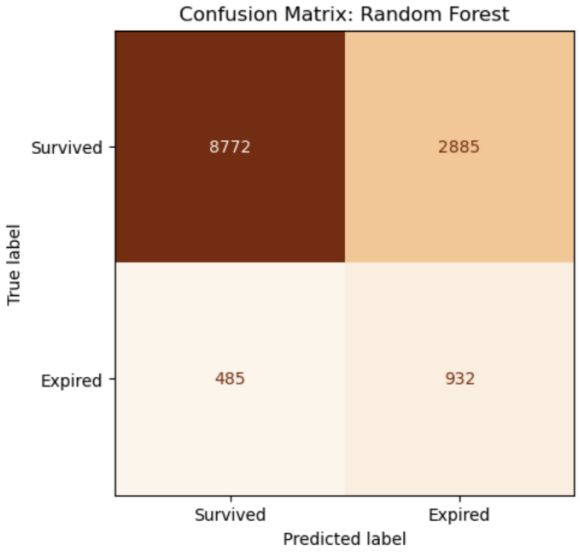
--- Decision Tree Classification Report ---
              precision    recall  f1-score   support

    Survived       0.95      0.73      0.82     11657
     Expired       0.24      0.72      0.36      1417

    accuracy                           0.72     13074
   macro avg       0.60      0.72      0.59     13074
weighted avg       0.88      0.72      0.77     13074
```

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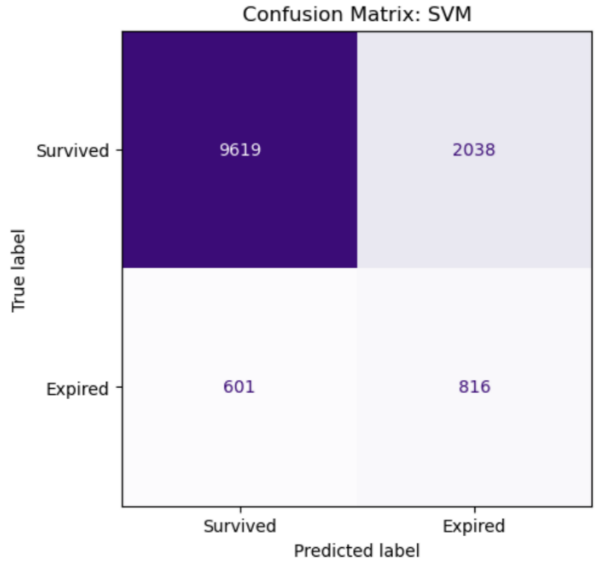
--- Random Forest Classification Report ---
              precision    recall  f1-score   support

    Survived       0.95      0.75      0.84     11657
     Expired       0.24      0.66      0.36      1417

    accuracy                           0.74     13074
   macro avg       0.60      0.71      0.60     13074
weighted avg       0.87      0.74      0.79     13074
```

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--- SVM Classification Report ---
              precision    recall  f1-score   support

    Survived       0.94      0.83      0.88     11657
     Expired       0.29      0.58      0.38      1417

    accuracy                           0.80     13074
   macro avg       0.61      0.70      0.63     13074
weighted avg       0.87      0.80      0.83     13074
```

Figure 4: Confusion Matrix and Classification Report (Logistic Regression, Decision Tree, Random Forest, SVM)

6. Discussion

Only three variables were retained after L1 regularization: SpO_2 minimum, GCS motor score, and GCS eye response. This suggests that a small subset of physiological indicators may already capture most of the predictive information related to ICU mortality. From a clinical perspective, this result is reasonable. Oxygen saturation reflects respiratory function, while the Glasgow Coma Scale evaluates neurological status. These two indicators are fundamental indicators of patient deterioration in critical care settings. This may explain why models using only these features were still able to achieve reasonable predictive performance.

The simplified model achieved an AUC of around 0.77, compared with 0.87 for the model using all features. Although models trained with the full feature set achieved higher AUC values, the simplified model may still be valuable in practice due to its interpretability and lower data requirements. Using only a small number of variables may allow faster risk assessment in real clinical environments.

Among the models tested, the Decision Tree achieved the best balance between performance and recall. This may be because clinical decision processes often follow rule-based patterns, which are naturally captured by decision tree models.

Some limitations should be acknowledged. For example, categorical variables such as race were encoded directly without additional grouping. In future work, category consolidation could reduce sparsity. In addition, this study used only static measurements rather than time-series data.

Future studies could incorporate temporal patterns in vital signs or apply more advanced models designed for sequential clinical data.

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7. Conclusion

This study developed machine learning models to predict ICU mortality risk using the MIMIC dataset. After preprocessing and feature selection using L1-regularized logistic regression, three key predictors were identified: minimum oxygen saturation, GCS motor score, and GCS eye response.

Among the evaluated models, the Decision Tree achieved the best performance when using the selected features. Although models trained with the full feature set showed higher AUC, the simplified model demonstrated that a small number of clinically meaningful variables can still provide useful predictions.

Overall, the results illustrate how feature selection can reduce model complexity while retaining reasonable predictive performance.

8. AI Tool Declaration

AI-based tools were used to assist with code clarification, troubleshooting programming issues, and improving the wording of the written report. All analyses, model implementation, and interpretation of results were conducted by the author.